California Housing Price Prediction – Project Report

Summary

The model selected by the project (a plain linear regression) predicts a district's average house price with a mean error of **about $53,320**, and explains **57.6%** of the variation in prices on data it had never seen ($R^2 = 0.576$).

- Against the naive baseline (always predict the average price): about $90,600 error and $R^2 = 0$, so the model is clearly better.
- The most important factor is the district's median income (MedInc), with a correlation of 0.688 with price.
- The more complex models (polynomial degree 3 and 5) collapsed on the test data because of extreme values in a few districts. A verification run on a public copy of the same data suggests that limiting extreme values would let polynomial models reach R^2 of about 0.73–0.75, and tree-based models about 0.80–0.83 (indicative; see "How these results were checked").
- The whole project ran end to end on the real California Housing data (20,640 districts) in about 107 seconds.

In short: the linear model was selected because it is the most **stable**, not because it is the most accurate. Clearly better results are within reach (see the next steps).

Data and train/test split

The project uses the **California Housing** dataset that ships with scikit-learn: **20,640 districts** and **8 features**, with no missing values. The target is the district's median house value in units of $100,000 (mean 2.069, i.e. about $206,900).

The data was split 80% for training (16,512 rows) and 20% for testing (4,128 rows). The test data is never used for training or for choosing the model; it is used once, at the end.

Note: each row is a **residential district** (block group) from the 1990 California census, not a single house. The predicted price is therefore the **district's average house price**, and the word "house" in this report means such a district.

Feature	Meaning	Mean	Correlation with price
MedInc	Median income of the district	3.87	0.688

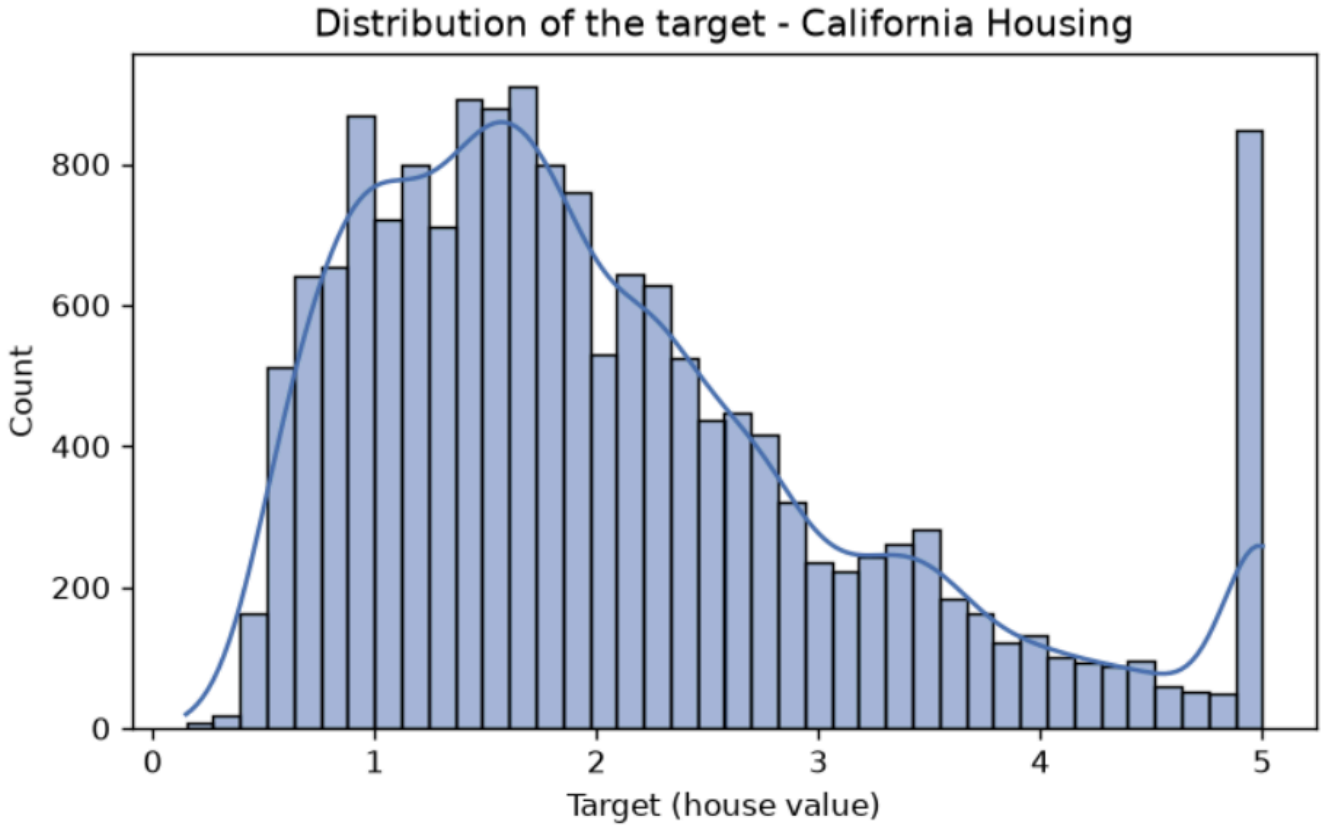
Feature	Meaning	Mean	Correlation with price
AveRooms	Average number of rooms	5.43	0.152
HouseAge	Median house age (years)	28.64	0.106
AveOccup	Average occupants per household	3.07	-0.024
Population	District population	1,425	-0.025
Longitude	Longitude	-119.57	-0.046
AveBedrms	Average number of bedrooms	1.10	-0.047
Latitude	Latitude	35.63	-0.144

Data caveats: the highest recorded price is 5.0 ($500,000) and values are capped there (about 5% of districts sit at this cap). There are also extreme districts, for example AveOccup reaches 1,243.

Exploring the data with figures

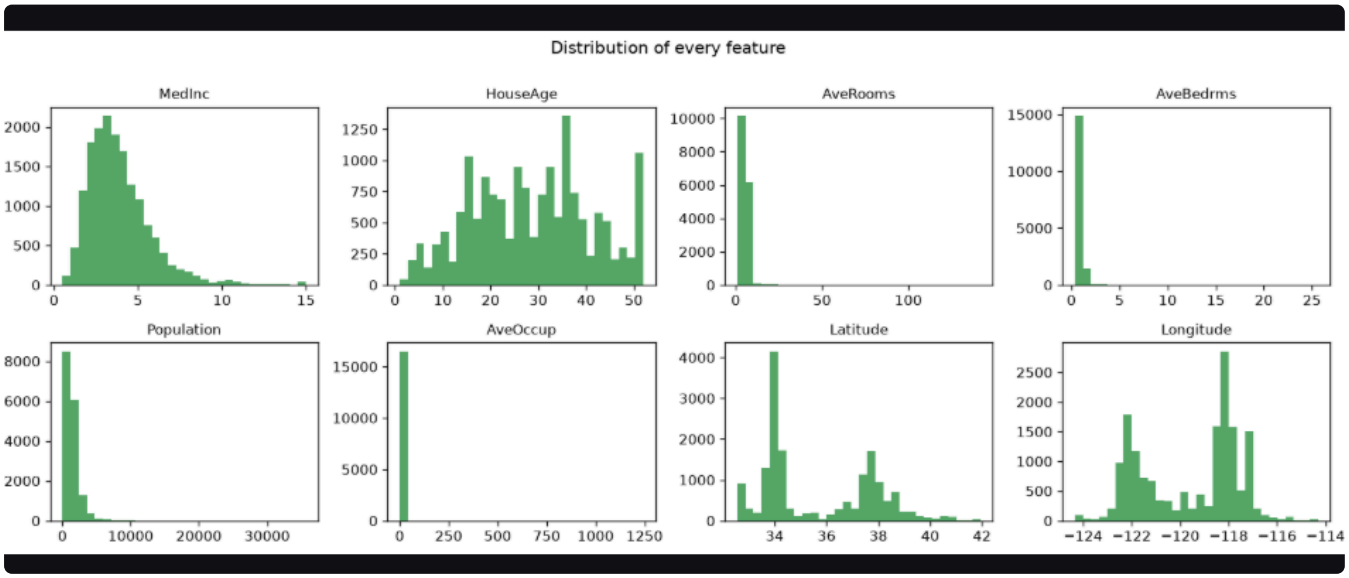
The five figures below are the project's actual output (the `outputs/figures` folder). They describe the data before any model is built.

Figure 1: Distribution of prices



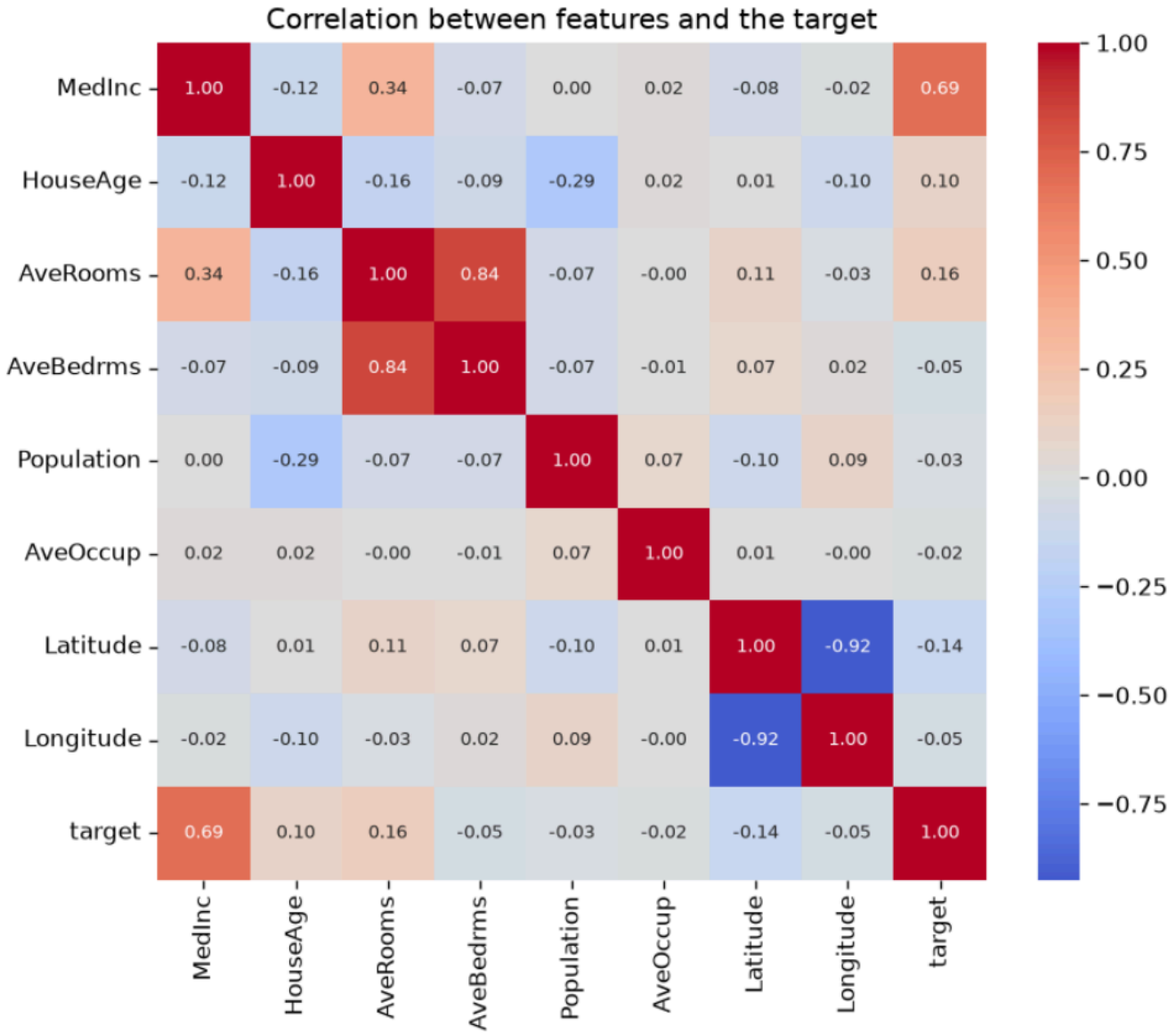
Most districts are priced between 1 and 3 (that is, $100,000 to $300,000), with a long tail toward higher prices. The striking feature is the **sharp spike at 5.0**: about 5% of all districts sit at the maximum value itself. This number is capped in the data and is not a real price.

Figure 2: Distribution of the features



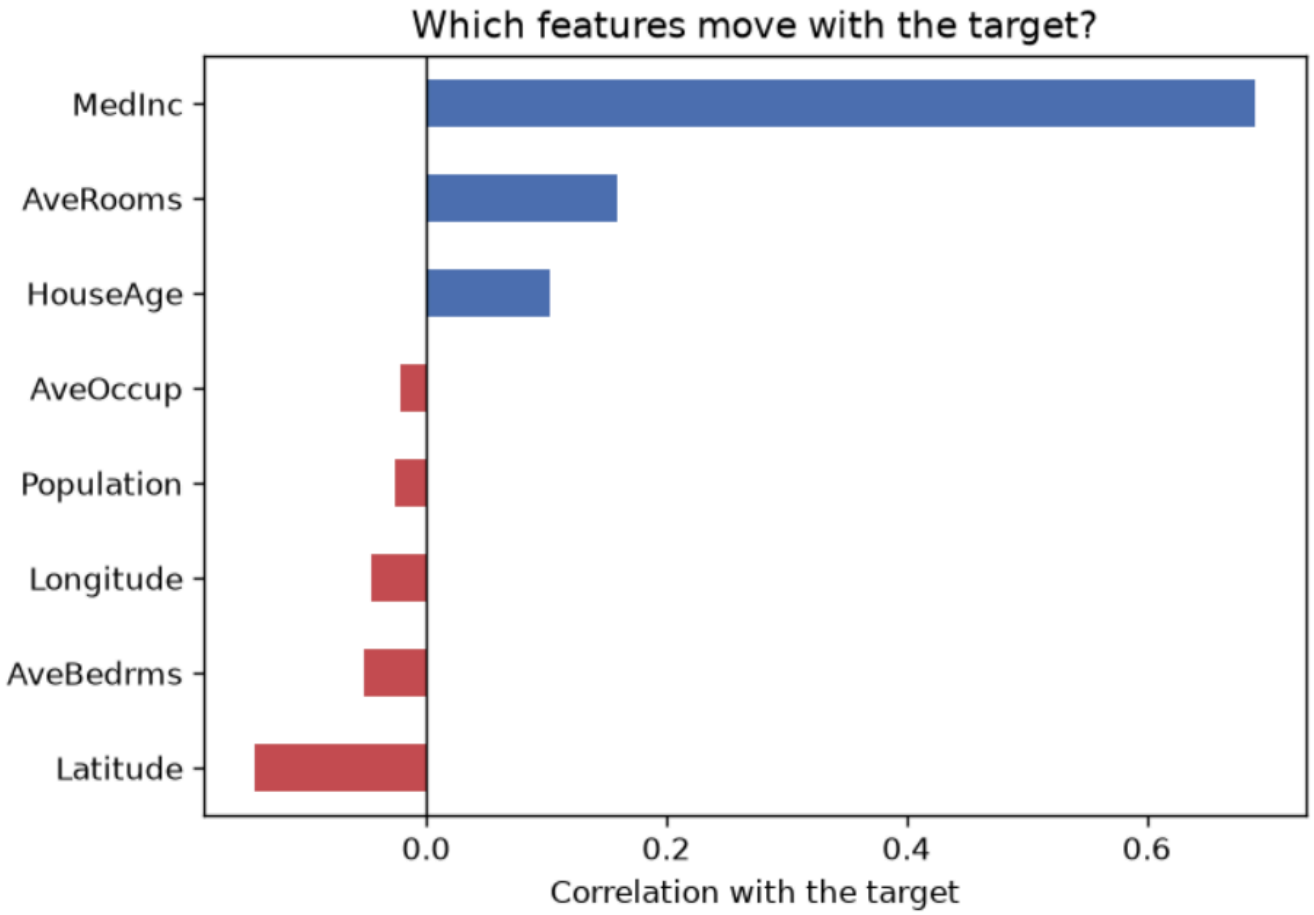
- Median income (MedInc) is skewed toward small values, mostly between 2 and 5.
- AveRooms, AveBedrms, Population and AveOccup look like a single thin bar near zero because they contain **extreme values** (AveRooms up to about 141, AveOccup up to 1,243, Population up to 35,682). These values are the reason the complex models fail later.
- Latitude and longitude each show two clear peaks, because districts cluster around two large areas (southern and northern California).

Figure 3: Correlation matrix



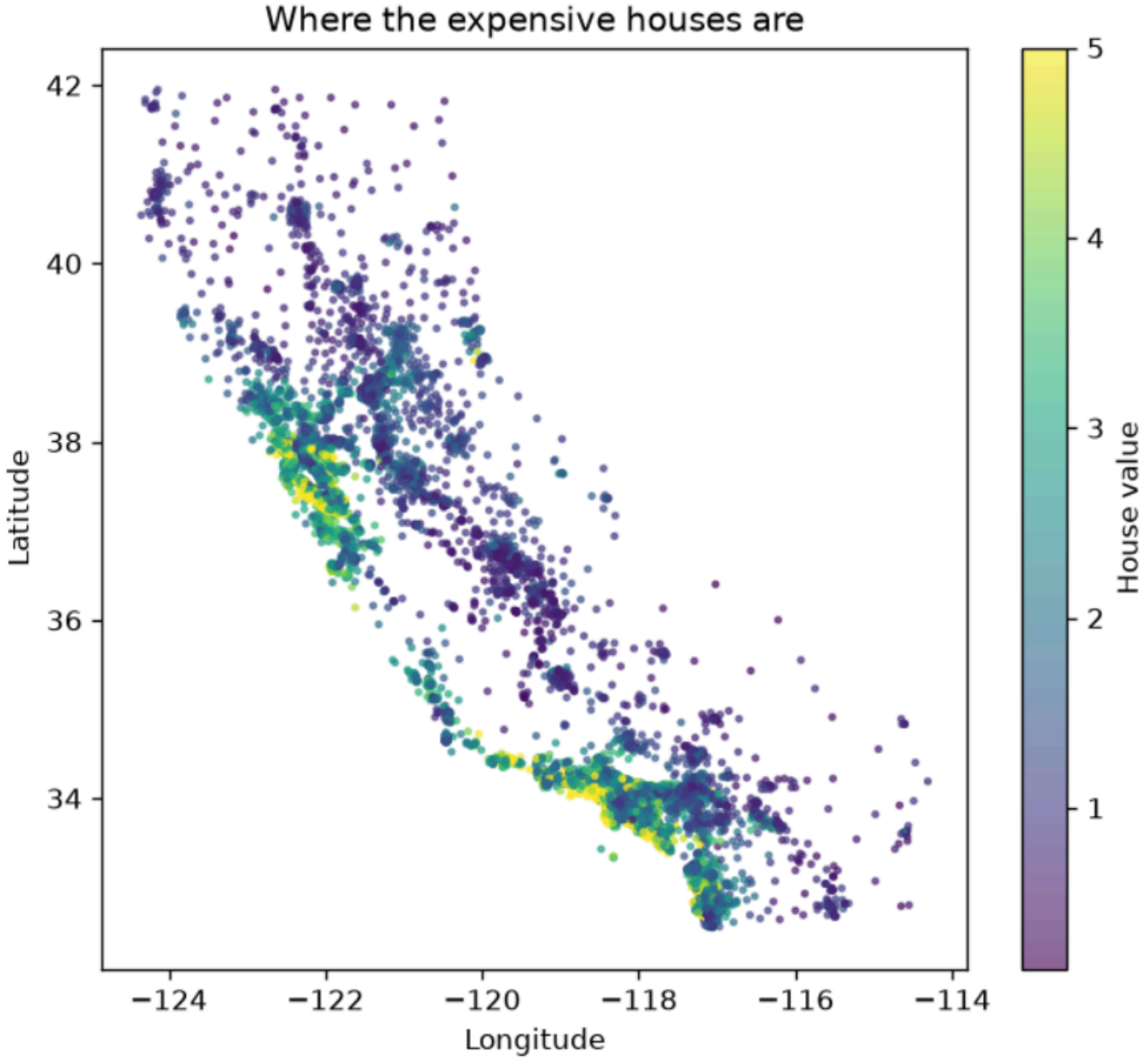
The strongest correlation between two features is between latitude and longitude (**-0.92**), followed by average rooms and average bedrooms (**0.84**). In each pair the two features carry overlapping information, which affects how the model's coefficients can be read (Figure 7).

Figure 4: Correlation of each feature with the price



Median income is by far the strongest feature, with a correlation of **0.69**, followed by number of rooms (0.15) and house age (0.11). The remaining features are weak (below 0.15 in absolute value).

Figure 5: Map of prices



The expensive districts (yellow) concentrate on the **coast**, especially around the San Francisco Bay (latitude about 37–38) and the Los Angeles and San Diego area (about 33–34), while the inland districts are cheaper (purple). Location is therefore an important factor, but a linear model can only use latitude and longitude as straight lines, so it does not fully capture this geographic pattern (an inference).

Models and comparison

Five models were tested and compared in three ways: on the training data, on the test data, and with 5-fold cross-validation (CV) on the training data only. R^2 = 1 means perfect prediction and 0 means no better than guessing the average.

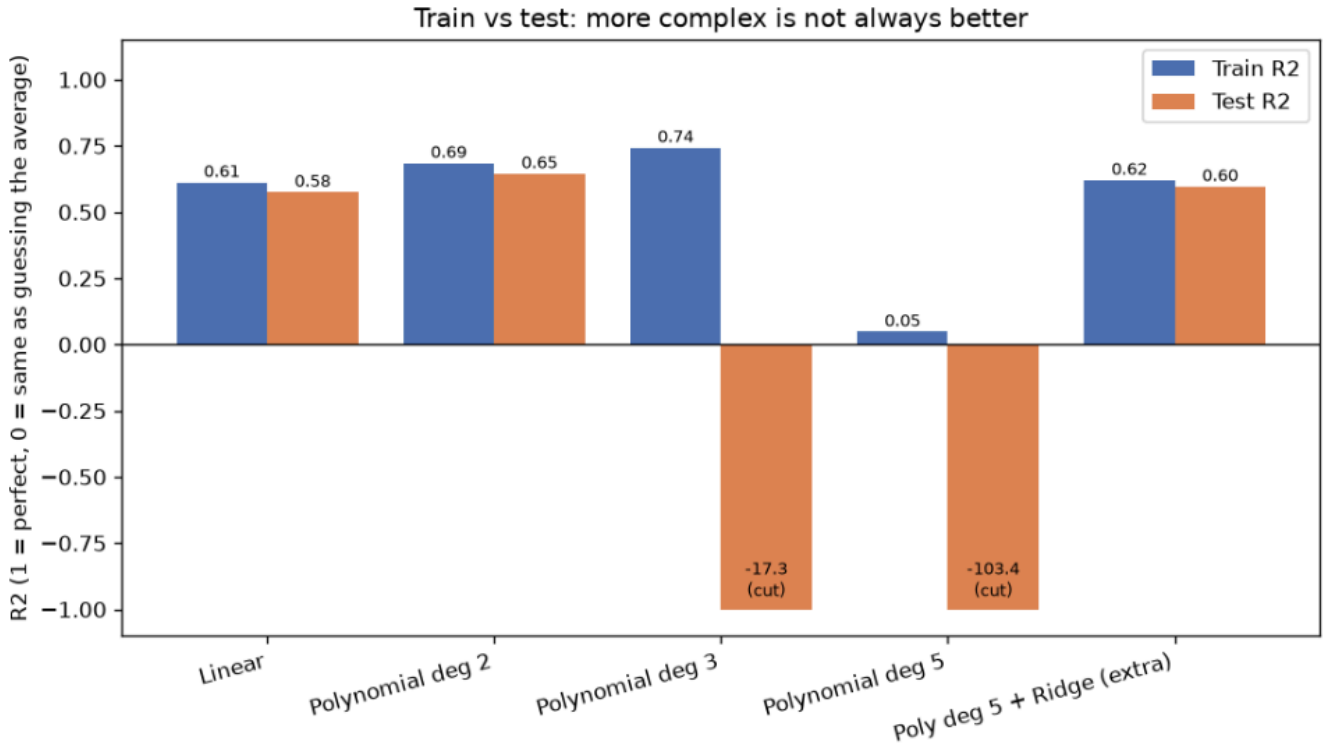
Model	Train R^2	Test R^2	CV R^2	Test MAE
Baseline (predict the mean)	—	-0.0002	—	0.906
Linear	0.6126	0.5758	0.6115	0.533
Polynomial degree 2	0.6853	0.6457	-1.8847	0.467
Polynomial degree 3	0.7439	-17.2689	-294,528.66	0.533
Polynomial degree 5	0.0487	-103.3509	-1,710.75	1.067
Degree 5 with Ridge (extra)	0.6214	0.5970	-206.52	0.527

MAE is in units of $100,000, so 0.533 is about $53,300.

What do these numbers mean?

- Degrees 3 and 5 failed on the test data (large negative R^2). This is the classic warning about complex models without regularization.
- Degree 5 scored $R^2 = 0.05$ even on the training data, which is not ordinary overfitting. The cause is extreme values. In the verification run on a public copy of the same census data, the same degree-5 model again scored a training R^2 of about 0.05, and when each feature was capped at its 1st–99th percentile the training R^2 rose to 0.83. Raising extreme values to the 5th power produces enormous numbers, and the least-squares solver loses precision.
- The Ridge model addresses this problem and reached $R^2 = 0.597$ on the test data, slightly above the linear model. Its CV average is negative because of a single bad fold: in the verification run four of the five folds scored between 0.55 and 0.59 and one fold scored -3.46.
- Degree 2 scored best on the test data ($R^2 = 0.646$, error about $46,700), but its CV score is poor (-1.88). The reason is that two of the five folds failed (-0.59 and -1.62 in the verification run) while the other three scored 0.55–0.67. Because the model was chosen by CV rather than by test results, the choice settled on the linear model, the only one that was stable in every fold.

Figure 8: Train vs test for each model



Blue bars are training and orange bars are test. Only three models succeeded on the test data: linear, degree 2 and Ridge, and their results are close (the gap between them is under 0.07). Degree 3 and degree 5 show large negative values (-17.3 and -103.4) that were cut off at -1 in the chart so the rest stays readable. Note that degree 5 is not merely overfitting: it is weak even on the training data (0.05).

Final result and examples

The final model (linear regression) was saved and evaluated **once** on 4,128 districts it had never seen:

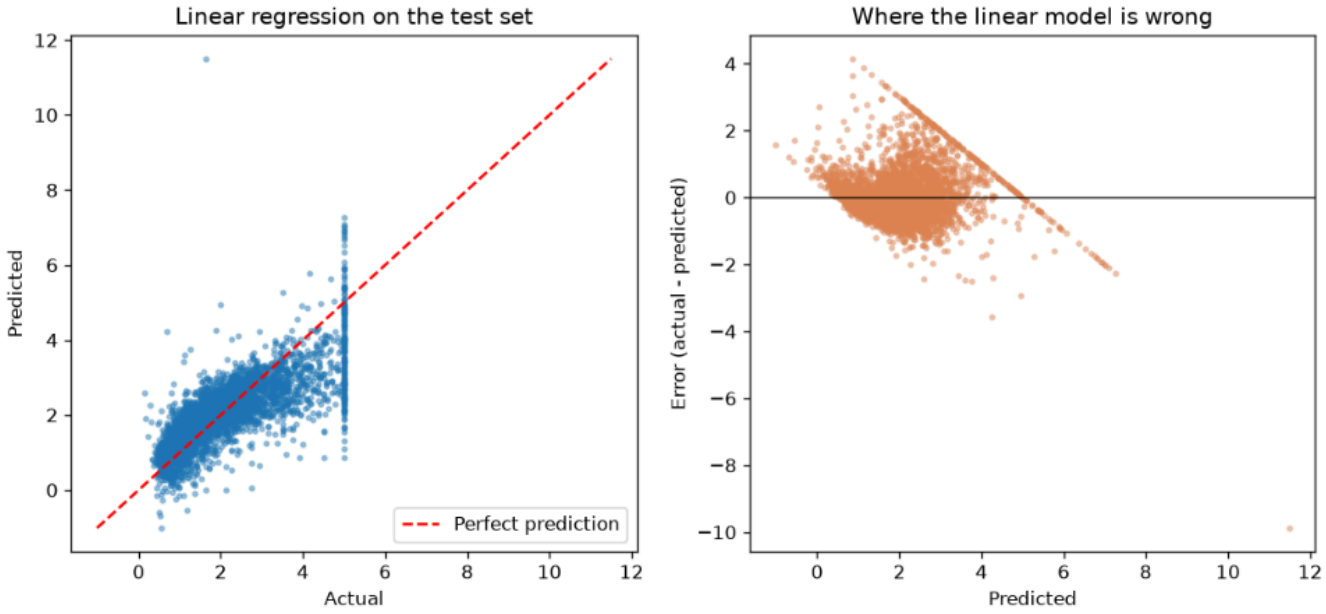
Metric	Value	Meaning
R^2	0.5758	Explains about 58% of the variation in prices
MAE	0.533	Average error of about $53,320 per district
RMSE	0.746	Root-mean-square error, penalizes large errors more (about $74,600)
MAPE	31.9%	Average relative error vs the true price
RMSPE	55.4%	Root-mean-square relative error (inflated by cheap districts)

A test on 5 districts the model had not seen (prices in dollars):

District #	Predicted	Actual
20046	71,912	47,700
3024	176,402	45,800
15663	270,966	500,001
20484	283,893	218,600
9814	260,466	278,000

The prediction is very close for district 9814, and far off for 3024 (about $130,000 too high) and 15663 (about $229,000 too low, because the true price is capped at 500,001).

Figure 6: Predicted vs actual



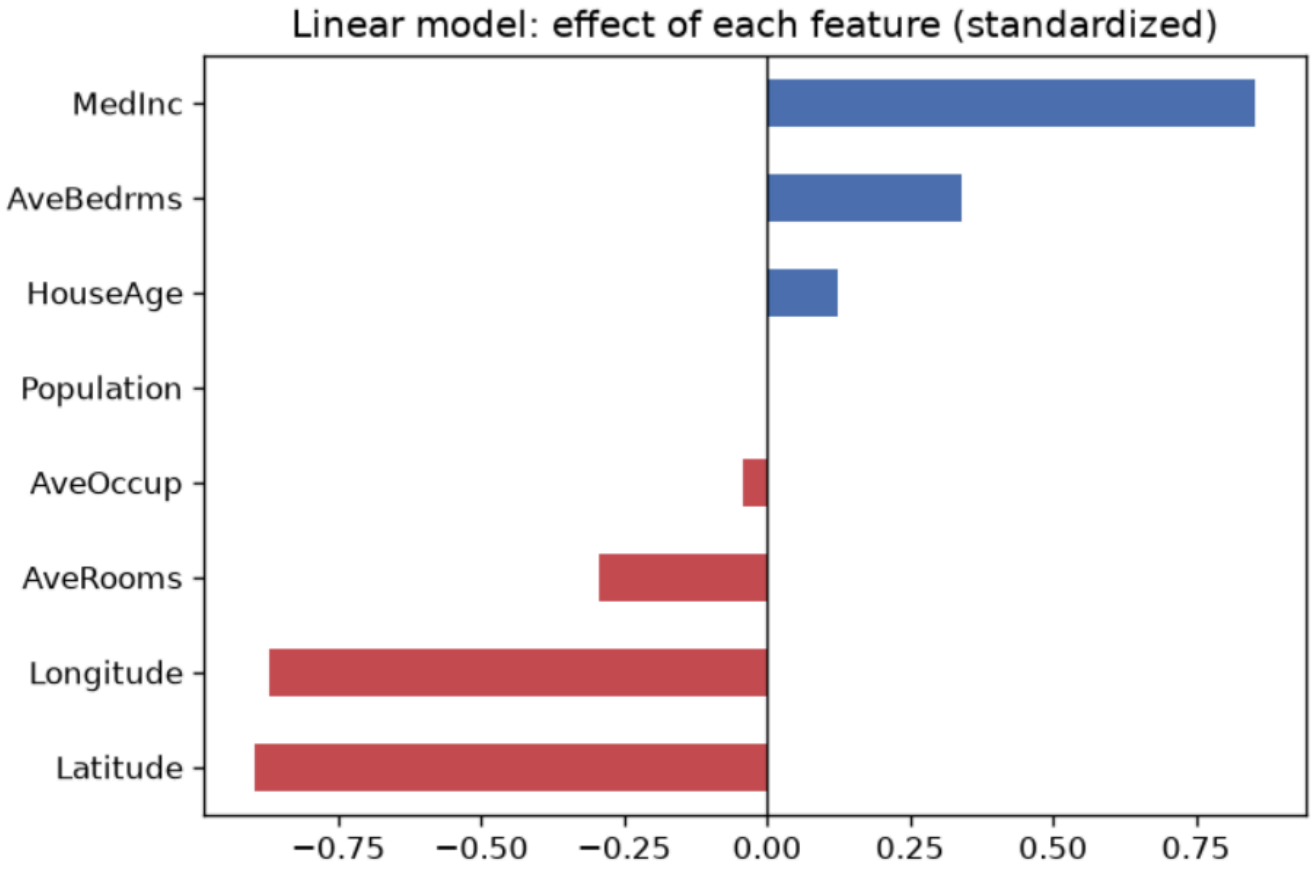
In the left plot the points follow the direction of the dashed line (perfect prediction) but are scattered, which is what $R^2 = 0.58$ looks like. Two things stand out:

- **A vertical column at the true price 5.0.** The capped districts are predicted at very different prices (from about 1 to over 7). This is a result of the cap at 5.0 in the data.
- **A diagonal line in the error plot (right).** These are the capped districts: every district whose true value is 5.0 has an error equal to 5 minus its prediction, which forms exactly such a line.

There is also a single extreme point predicted at about 11.5 (over $1 million) whose true value is about 1.7, and some predictions fall below zero, which are meaningless prices (20 of 4,086 test districts in the verification run). In that run the capped districts were 4.9%

of the test rows but 12.6% of the total absolute error, and their average prediction was 3.88 against a true value of 5.0.

Figure 7: The effect of each feature in the linear model



The bars compare each feature's effect after putting the features on the same scale. Median income (MedInc) is the strongest positive factor (about +0.85), while latitude and longitude are the strongest negative factors (about -0.9 each).

Important caution when reading it: the AveRooms coefficient is negative (about -0.29) even though its correlation with price is positive (0.15), and the AveBedrms coefficient is positive (about +0.33) even though its correlation is negative (-0.05). The reason is that these two features are correlated at 0.84, so the model splits the effect between them. The same applies to latitude and longitude (-0.92). Read these coefficients as a combined effect per pair, not as the independent effect of each feature.

Limitations and next steps

Limitations

- The average error of $53,320 is large relative to the average price (about $206,900). The model suits learning and demonstration, not real purchase decisions.

- The data is old (1990 census) and prices are capped at 500,001, so expensive districts are predicted lower than their price.
- The evaluation uses a single test split; results may change slightly with a different split.
- CV averages for the complex models are distorted by one or two folds that fail because of extreme districts, so an average CV score does not describe how those models behave on the other folds.
- The coefficients of the linear model describe statistical relationships, not causes, and they overlap because the features are correlated (Figures 3 and 7).

Next steps

1. Cap extreme feature values (for example at the 1st and 99th percentiles) and re-test polynomial degrees 2 and 3. In the verification run this lifted test R^2 from 0.59 (linear) to about 0.73 (degree 2) and 0.75 (degree 3).
2. Treat the capped price (5.0) separately, for example by excluding those districts, since in the verification run they caused about 13% of the linear model's error while being about 5% of the data.
3. Try tree-based models: in the verification run gradient boosting reached a test R^2 of 0.83 and Random Forest 0.80, against 0.59 for the linear model.
4. Compare the models on more than one split (repeated cross-validation) to confirm that the ranking is stable.

How these results were checked

Three kinds of evidence are used in this report, and each is marked so the reader knows how far to trust it:

Evidence	What it covers	How reliable
Project output	Every number in the tables (R^2, MAE, RMSE, MAPE, CV), the five example predictions, the counts of rows, and the figures themselves	Exact: copied from the project's own run on 20,640 districts
Reading the figures	Bar lengths and positions, such as the coefficients in Figure 7 and the hot spots on the map	Approximate
Verification run	The explanations for why the complex models fail, and the indicative results for improved models	Independent re-run on a public copy of the same census data; to be re-confirmed on the project's own data

The public copy has 20,433 districts (207 rows with missing bedroom counts are dropped) and its features were rebuilt the way scikit-learn defines them, so its numbers differ slightly from the project's. They agree closely where it matters: linear test R^2 0.587 (project: 0.576), degree-5 training R^2 0.050 (0.049), and the standardized linear coefficients match the pattern and size shown in Figure 7 (MedInc +0.84, AveBedrms +0.33, AveRooms -0.29, Latitude -0.90, Longitude -0.87). Source: public copy of the California Housing data.

Appendix: how the project works

The whole project is one Python script that runs end to end in a single click (about 107 seconds on the real data):

1. **Load and inspect** the data: shape, missing values, summary statistics, correlations.
2. **Split** into 80% training and 20% test.
3. **Explore** with figures (Figures 1–5).
4. **Baseline**: always predict the average price. Any model must beat it.
5. **Compare models** (linear, polynomial degrees 2, 3 and 5, and degree 5 with Ridge). Each model is a single pipeline, so feature scaling is learned from the training part only.
6. **Choose** the best model by cross-validation on the training data, save it, and evaluate it **once** on the test data.
7. **Predict** on districts the model has not seen, and write an HTML report with all numbers and figures.

Glossary of the metrics

Term	Plain meaning
R^2	Share of price variation the model explains. 1 is perfect, 0 equals guessing the average, negative is worse than guessing
MAE	Average absolute error in price units (here $100,000)
RMSE	Like MAE but penalizes large errors more
MAPE / RMSPE	The same errors expressed as a percentage of the true price
Cross-validation (CV)	Splitting the training data into 5 parts and testing on each part in turn, for a more reliable estimate

Term	Plain meaning
Overfitting	A model that fits the training data very well but fails on new data

Data source: the California Housing dataset provided by scikit-learn (1990 U.S. census, downloaded automatically on the first run).